



ELECTRO MAGNETI C THEORY(EMFT)

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and

El ect r oni cs Engi neer i ng

ELECTRO MAGNETI C FI ELD

THEORY

Course I nt r oduct i on

Dr. Venugopal . N

Department of Electrical and
Electronics Engineering

ELECTRO MAGNETIC FIELD THEORY

Course Introduction



Electromagnetic theory is a branch of physics or electrical engineering which involves

The study of electric and magnetic effects

Electromagnetic force

Electromagnetic fields.

ELECTRO MAGNETIC THEORY

Electromagnetic Fields significance

The electromagnetic fields are
used in

Induction heaters

Electric heating equipment

Agriculture and Medicine

ELECTRO MAGNETIC THEORY

Applications of Electromagnetic Theory

Some of the electromagnetic devices most frequently used are,

Transformers

Electric Relays

Transmission Lines

Electric Motors

ELECTRO MAGNETIC THEORY

Applications of Electromagnetic Theory Contd. ,

Radi o/ TV

Opt i cal
f i ber s

Tel ephone

Radar s

Ant ennas

Laser s
et c. ,

ELECTRO MAGNETIC THEORY



Applications of Electro Magnetic

Principles

Electro Magnetic principles find applications in various disciplines such as electrical Machines

o Magnetic Interference and Compatibility

uclear research

Fiber optics

ELECTRO MAGNETIC THEORY



Applications of Electromagnetic

Principles Contd.,

Microwaves

Antennas

as

Satellite

Communications

Electromechanical energy

conversion

Radar meteorology and

Remote sensing.

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Course Content



The electromagnetic theory course is divided into five units
Unit 1: Electrostatics

2: Electric Field in material space

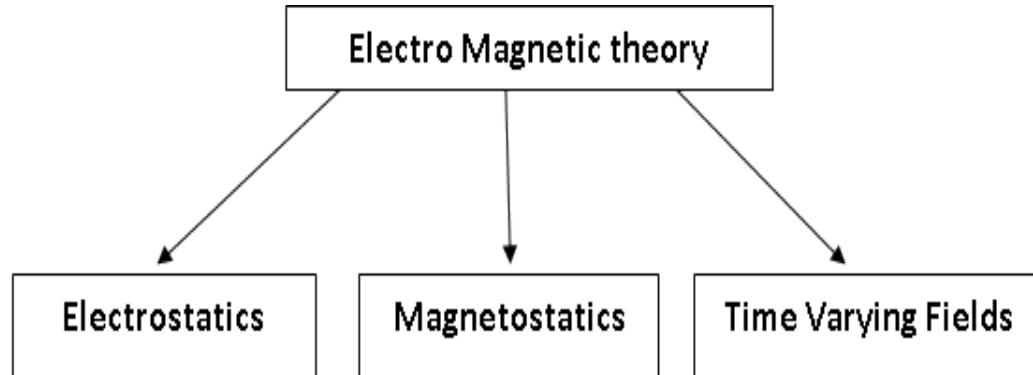
Unit 3:

Magnetostatics

Unit 4: Maxwell's Equation and Time Varying Fields

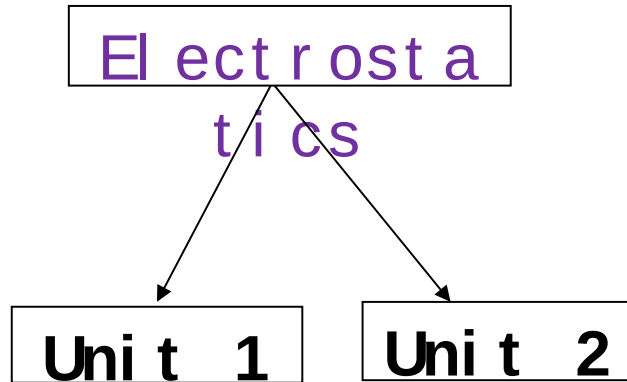
ELECTRO MAGNETIC THEORY

Course Overview



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Electrostatics



ELECTRO MAGNETIC THEORY

Magnetostatics

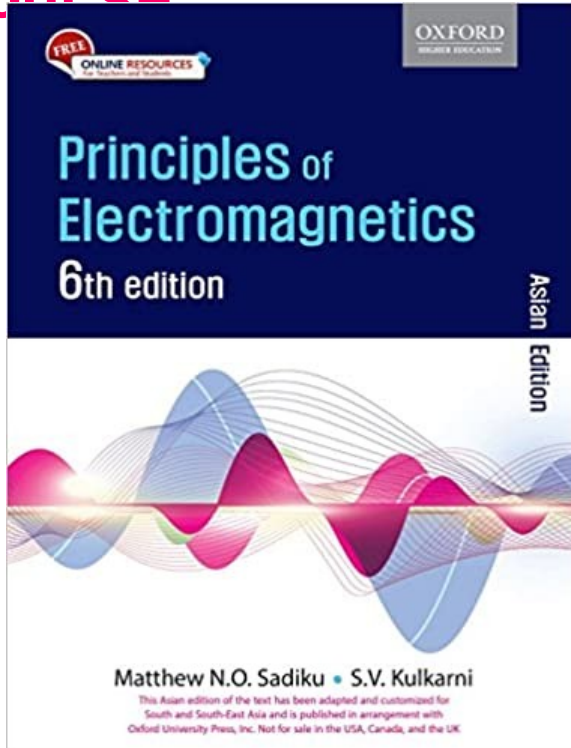
Magnetostatics, Maxwell's Equations
and Time Varying Fields

Unit 3

Unit 4

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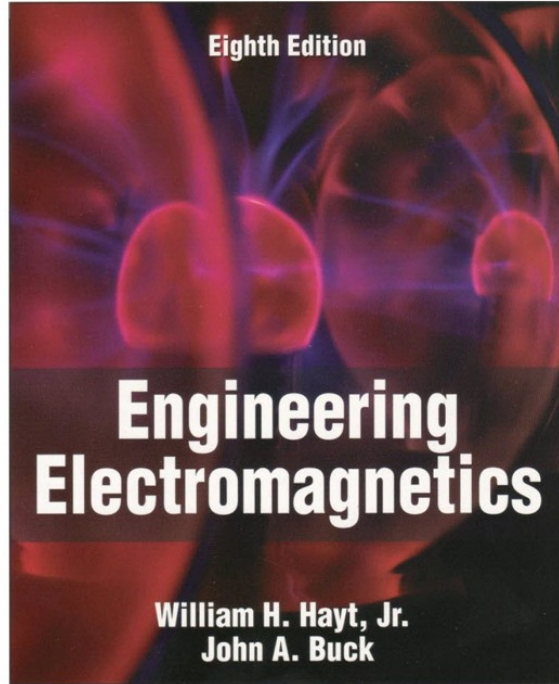
Prescribed Text Book for the
course



“Principles of Electromagnetics”, Matthew N. O. Sadiku and S. V. Kulkarni, (2015 Sixth Edition), Oxford University

ELECTRO MAGNETIC THEORY

Prescribed Text Book for the



“Engineering
Electromagnetics”, William H.
Hayt, Jr. John
A. Buck
(2020 Eighth
Edition),
McGraw Hill
International
Edition,

ELECTRO MAGNETIC FIELD THEORY

Vectors Algebra



1. Vector Addition

$$\vec{A} + \vec{B} = \vec{B} + \vec{A} \quad \text{Commutative law}$$

$$\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C} \quad \text{Associative law}$$

2. Vector Subtraction

$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$$

3. Multiplication by a scalar

$$m(\vec{A}) = \vec{A}m \quad \text{Commutative law}$$

$$m(n\vec{A}) = n(m\vec{A}) \quad \text{Associative law}$$

$$(m+n)\vec{A} = m\vec{A} + n\vec{A} \quad \text{Distributive law}$$

$$m(\vec{A} + \vec{B}) = m\vec{A} + m\vec{B} \quad \text{Distributive law}$$

ELECTRO MAGNETIC THEORY

Unit Vector

A unit vector $\mathbf{a}_A$ along $\mathbf{A}$ is defined as a vector whose magnitude is unity (i.e., 1) and its direction is along $\mathbf{A}$, that

$$\mathbf{a}_A = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{\mathbf{A}}{A}$$

- Scalar notation: A or A
(*italic* or plain)
- Vector notation: $\mathbf{A}$ or $\mathbf{A}$ (**bold**
or plain with arrow)

ELECTRO MAGNETIC THEORY

Unit Vector Contd. ,



A vector **A** in Cartesian (or rectangular) coordinates may be represented (A_x, A_y, A_z) or $A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z$

The magnitude of vector **A** is given by

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

The unit vector along **A** is given by

$$\mathbf{a}_A = \frac{A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}}$$

ELECTRO MAGNETIC THEORY

Dot Product

Given two vectors A and B , the dot product, or scalar product, is defined as $A \cdot B = |A||B|\cos\theta_{AB}$

The dot product is a scalar, and it obeys the commutative law.

$$A \cdot B = B \cdot A$$

ELECTRO MAGNETIC THEORY

Dot Product Contd. ,



For any vector $\mathbf{A} = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z$ and
 $\mathbf{B} = B_x \mathbf{a}_x + B_y \mathbf{a}_y + B_z \mathbf{a}_z$

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z$$

Given $\mathbf{A} = 2\mathbf{a}_x - 3\mathbf{a}_y + \mathbf{a}_z$ and $\mathbf{B} = -4\mathbf{a}_x - 2\mathbf{a}_y + 5\mathbf{a}_z$,

find $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$

$$\begin{aligned}\mathbf{A} \cdot \mathbf{B} &= A_x B_x + A_y B_y + A_z B_z \\ &= 2(-4) - 3(-2) \\ &\quad + 1(5) \\ &= -8 + 6 + 5\end{aligned}$$

ELECTRO MAGNETIC THEORY

Cross Product



$$\vec{C} = \vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin \theta \hat{n}$$

Ex.,

where ' θ ' is angle between $\vec{A}$ and $\vec{B}$ ($0 \leq \theta \leq \pi$),

and $\hat{n}$ is unit vector perpendicular to plane of $\vec{A}$ and $\vec{B}$

directed such that $\vec{A}$ $\vec{B}$ $\vec{C}$ form a right handed system of vectors

$$\vec{A} \times \vec{B} = - \vec{B} \times \vec{A}$$

$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$$

ELECTRO MAGNETIC THEORY

Cross Product

Cont d. ,

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$= (A_y B_z - A_z B_y) \mathbf{a}_x + (A_z B_x - A_x B_z) \mathbf{a}_y + (A_x B_y - A_y B_x) \mathbf{a}_z$$

ELECTRO MAGNETIC THEORY

Cross Product Contd. ,



Given $\mathbf{A} = 2\mathbf{a}_x - 3\mathbf{a}_y + \mathbf{a}_z$ and $\mathbf{B} = -4\mathbf{a}_x - 2\mathbf{a}_y + 5\mathbf{a}_z$, find $\mathbf{A} \times \mathbf{B}$.

$$\mathbf{A} \times \mathbf{B} = (A_y B_z - A_z B_y)\mathbf{a}_x + (A_z B_x - A_x B_z)\mathbf{a}_y + (A_x B_y - A_y B_x)\mathbf{a}_z$$

$$= ((-3)(5) - (1)(-2))\mathbf{a}_x + ((1)(-4) - (2)(5))\mathbf{a}_y + ((2)(-2) - (-3)(-4))\mathbf{a}_z$$

$$= \underline{\underline{-13\mathbf{a}_x - 14\mathbf{a}_y - 16\mathbf{a}_z}}$$

ELECTRO MAGNETIC THEORY

Dot and Cross Product of Unit

Vectors

$$\mathbf{a}_x \cdot \mathbf{a}_x = \mathbf{a}_y \cdot \mathbf{a}_y = \mathbf{a}_z \cdot \mathbf{a}_z = 1.$$

$$\mathbf{a}_x \cdot \mathbf{a}_y = \mathbf{a}_y \cdot \mathbf{a}_x = \mathbf{a}_x \cdot \mathbf{a}_z = \mathbf{a}_z \cdot \mathbf{a}_x = 0.$$

$$\mathbf{a}_x \times \mathbf{a}_x = \mathbf{a}_y \times \mathbf{a}_y = \mathbf{a}_z \times \mathbf{a}_z = 0.$$

$$\mathbf{a}_x \times \mathbf{a}_y = \mathbf{a}_z$$

$$\mathbf{a}_y \times \mathbf{a}_z = \mathbf{a}_x$$

$$\mathbf{a}_z \times \mathbf{a}_x = \mathbf{a}_y$$

$$\mathbf{a}_y \times \mathbf{a}_x = -\mathbf{a}_z$$

$$\text{Note } \mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$$

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ELECTRO MAGNETIC THEORY

Coordinate

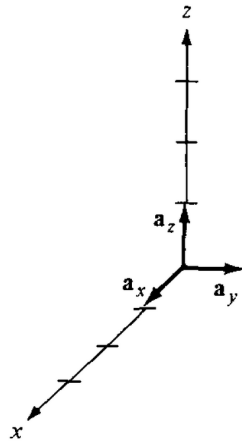
Systems

For an explicit representation of a vector quantity, a 'coordinate system' is essential.

Sl. No.	System	Co-ordinate variables	Unit vectors
1.	Rectangular	x, y, z	$\mathbf{a}_x, \mathbf{a}_y, \mathbf{a}_z$
2.	Cylindrical	ρ, ϕ, z	$\mathbf{a}_\rho, \mathbf{a}_\phi, \mathbf{a}_z$
3.	Spherical	r, θ, ϕ	$\mathbf{a}_r, \mathbf{a}_\theta, \mathbf{a}_\phi$

ELECTRO MAGNETIC THEORY

Rectangular Coordinate



$$-\infty < x < \infty$$

$$-\infty < y < \infty$$

$$-\infty < z < \infty$$

A vector $\mathbf{A}$ in Rectangular

(otherwise known as

Cartesian) coordinates can be

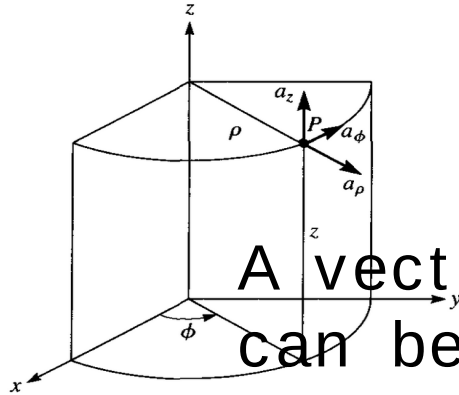
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$$(A_x, A_y, A_z) \quad \text{or} \quad A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z$$

Where $\mathbf{a}_x$, $\mathbf{a}_y$, and $\mathbf{a}_z$ are unit vectors along the x, y and z directions

ELECTRO MAGNETIC THEORY

Cylindrical Coordinate Systems



$$0 \leq \rho < \infty$$

$$0 \leq \phi < 2\pi$$

$$-\infty < z < \infty$$

A vector **A** in cylindrical coordinates can be written as

$$(A_\rho, A_\phi, A_z) \quad \text{or} \quad A_\rho \mathbf{a}_\rho + A_\phi \mathbf{a}_\phi + A_z \mathbf{a}_z$$

Magnitude of vector **A** can be written as

$$|\mathbf{A}| = (A_\rho^2 + A_\phi^2 + A_z^2)^{1/2}$$

ELECTRO MAGNETIC THEORY

Cylindrical Coordinate Systems

$$\mathbf{a}_\rho \cdot \mathbf{a}_\rho = \mathbf{a}_\phi \cdot \mathbf{a}_\phi = \mathbf{a}_z \cdot \mathbf{a}_z = 1$$

$$\mathbf{a}_\rho \cdot \mathbf{a}_\phi = \mathbf{a}_\phi \cdot \mathbf{a}_z = \mathbf{a}_z \cdot \mathbf{a}_\rho = 0$$

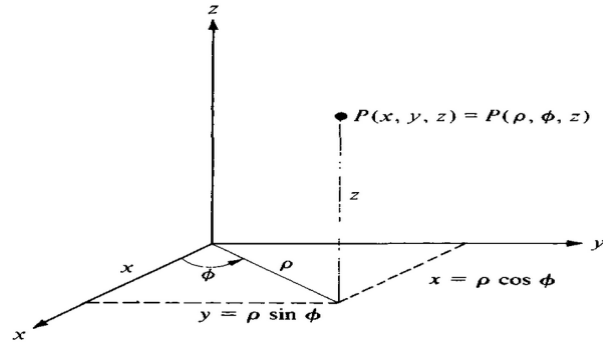
$$\mathbf{a}_\rho \times \mathbf{a}_\phi = \mathbf{a}_z$$

$$\mathbf{a}_\phi \times \mathbf{a}_z = \mathbf{a}_\rho$$

$$\mathbf{a}_z \times \mathbf{a}_\rho = \mathbf{a}_\phi$$

ELECTRO MAGNETIC THEORY

Cylindrical and Rectangular Coordinates Transformations



$$\rho = \sqrt{x^2 + y^2}, \quad \phi = \tan^{-1} \frac{y}{x}, \quad z = z$$

$$x = \rho \cos \phi, \quad y = \rho \sin \phi, \quad z = z$$

	$\mathbf{a}_\rho$	$\mathbf{a}_\phi$	$\mathbf{a}_z$
$\mathbf{a}_x \cdot$	$\cos \phi$	$-\sin \phi$	0
$\mathbf{a}_y \cdot$	$\sin \phi$	$\cos \phi$	0
$\mathbf{a}_z \cdot$	0	0	1

ELECTRO MAGNETIC THEORY

Cylindrical and Rectangular Coordinate Transformations



$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

The inverse of the transformation $(A_\rho, A_\phi, A_z) \rightarrow (A_x, A_y, A_z)$ is obtained as

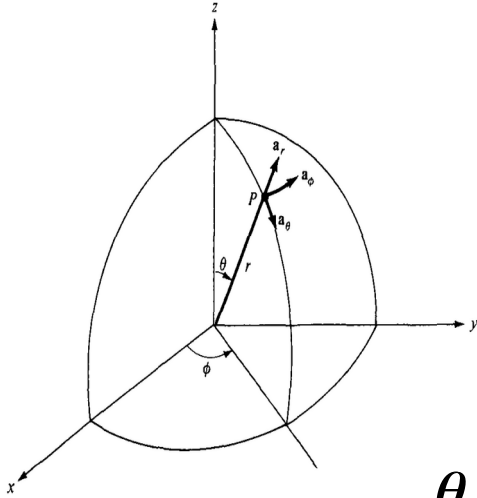
$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

or directly from eqs. (2.4) and (2.10). Thus

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

ELECTRO MAGNETIC THEORY

Spherical Coordinate System



$$r \geq 0, \quad 0 \leq \theta \leq \pi, \quad 0 \leq \phi < 2\pi$$

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$r = \sqrt{x^2 + y^2 + z^2}, \quad r \geq 0$$

$$\theta = \cos^{-1} \frac{z}{\sqrt{x^2 + y^2 + z^2}}, \quad 0^\circ \leq \theta \leq 180^\circ$$

$$\phi = \tan^{-1} \frac{y}{x}$$

ELECTRO MAGNETIC THEORY

Spherical Coordinate System

Contd. ,

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

	$\mathbf{a}_r$	$\mathbf{a}_\theta$	$\mathbf{a}_\phi$
$\mathbf{a}_x$	$\sin \theta \cos \phi$	$\cos \theta \cos \phi$	$-\sin \phi$
$\mathbf{a}_y$	$\sin \theta \sin \phi$	$\cos \theta \sin \phi$	$\cos \phi$
$\mathbf{a}_z$	$\cos \theta$	$-\sin \theta$	0

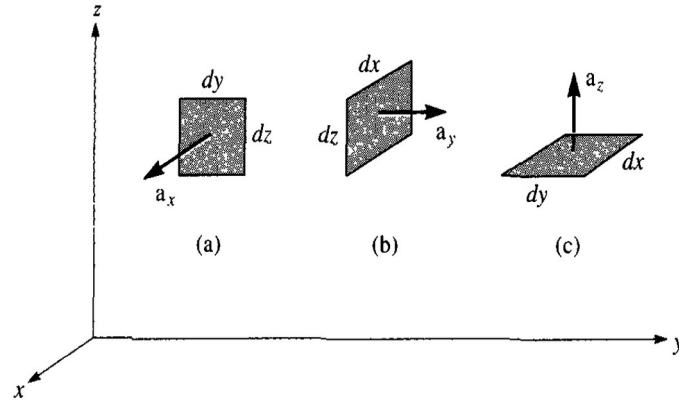
ELECTRO MAGNETIC THEORY

Differential Length, Differential Surface and Differential Volume

$$d\mathbf{l} = dx \mathbf{a}_x + dy \mathbf{a}_y + dz \mathbf{a}_z$$

$$d\mathbf{S} = \begin{matrix} dy \, dz \, \mathbf{a}_x \\ dx \, dz \, \mathbf{a}_y \\ dz \, dy \, \mathbf{a}_z \end{matrix}$$

$$dv = dx \, dy \, dz$$



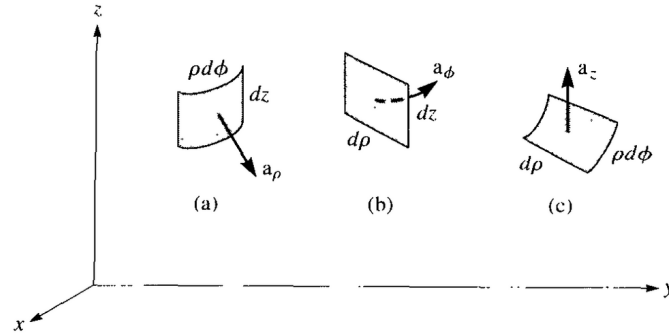
ELECTRO MAGNETIC THEORY

Differential Length, Differential Surface and Differential Volume

$$dl = d\rho \mathbf{a}_\rho + \rho d\phi \mathbf{a}_\phi + dz \mathbf{a}_z$$

$$d\mathbf{S} = \rho d\phi dz \mathbf{a}_\rho + d\rho dz \mathbf{a}_\phi + \rho d\phi d\rho \mathbf{a}_z$$

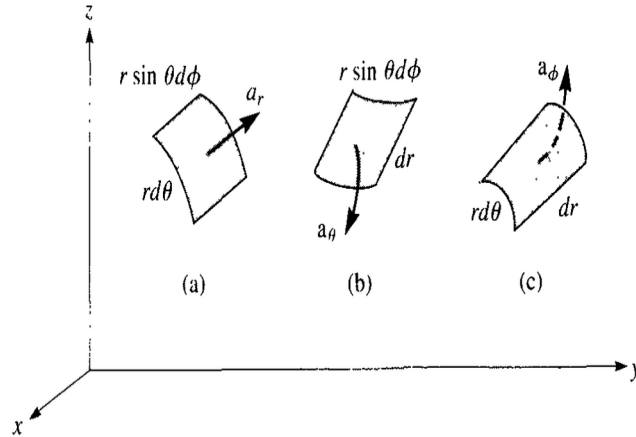
$$dv = \rho d\rho d\phi dz$$



ELECTRO MAGNETIC THEORY

Differential Length, Differential Surface and Differential Volume

$$d\mathbf{l} = dr \mathbf{a}_r + r d\theta \mathbf{a}_\theta + r \sin \theta d\phi \mathbf{a}_\phi$$



$$d\mathbf{S} = r^2 \sin \theta d\theta d\phi \mathbf{a}_r \\ r \sin \theta dr d\phi \mathbf{a}_\theta \\ r dr d\theta \mathbf{a}_\phi$$

$$dv = r^2 \sin \theta dr d\theta d\phi$$

ELECTRO MAGNETIC THEORY



Differential Length, Differential Surface and Differential Volume

- Using the differential length dl , find the length of each of the following curves:
 - $\rho = 3, \pi/4 < \phi < \pi/2, z = \text{constant}$
 - $r = 1, \theta = 30^\circ, 0 < \phi < 60^\circ$
 - $r = 4, 30^\circ < \theta < 90^\circ, \phi = \text{constant}$
- Calculate the areas of the following surfaces using the differential surface area dS :
 - $\rho = 2, 0 < z < 5, \pi/3 < \phi < \pi/2$
 - $z = 1, 1 < \rho < 3, 0 < \phi < \pi/4$
 - $r = 10, \pi/4 < \theta < 2\pi/3, 0 < \phi < 2\pi$
 - $0 < r < 4, 60^\circ < \theta < 90^\circ, \phi = \text{constant}$

ELECTRO MAGNETIC THEORY



Differential Length, Differential Surface and

Differential Volume (Assignment)

3. Use the differential volume dv to determine the volumes of the following regions:

(a) $0 < x < 1, 1 < y < 2, -3 < z < 3$

(b) $2 < \rho < 5, \pi/3 < \phi < \pi, -1 < z < 4$

(c) $1 < r < 3, \pi/2 < \theta < 2\pi/3, \pi/6 < \phi < \pi/2$

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ELECTRO MAGNETIC THEORY

Coordinate

Systems



Given point $P(-2, 6, 3)$ and vector $A = ya_x + (x + z)a_y$, express P and A in cylindrical and spherical coordinates. Evaluate A at P in the Cartesian, cylindrical, and spherical systems.

ELECTRO MAGNETIC THEORY

Coordinate

At point P : $x = -2$, $y = 6$, $z = 3$. Hence,

$$\rho = \sqrt{x^2 + y^2} = \sqrt{4 + 36} = 6.32$$

$$\phi = \tan^{-1} \frac{y}{x} = \tan^{-1} \frac{6}{-2} = 108.43^\circ$$

$$z = 3$$

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{4 + 36 + 9} = 7$$

$$\theta = \tan^{-1} \frac{\sqrt{x^2 + y^2}}{z} = \tan^{-1} \frac{\sqrt{40}}{3} = 64.62^\circ$$

Thus,

$$P(-2, 6, 3) = P(6.32, 108.43^\circ, 3) = P(7, 64.62^\circ, 108.43^\circ)$$

In the Cartesian system, $\mathbf{A}$ at P is

$$\mathbf{A} = 6\mathbf{a}_x + \mathbf{a}_y$$

ELECTRO MAGNETIC THEORY

Coordinate

For vector $\mathbf{A}$, $A_x = y$, $A_y = x + z$, $A_z = 0$. Hence, in the cylindrical system

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y \\ x + z \\ 0 \end{bmatrix}$$

or

$$A_\rho = y \cos \phi + (x + z) \sin \phi$$

$$A_\phi = -y \sin \phi + (x + z) \cos \phi$$

$$A_z = 0$$

But $x = \rho \cos \phi$, $y = \rho \sin \phi$, and substituting these yields

$$\begin{aligned} \mathbf{A} = (A_\rho, A_\phi, A_z) &= [\rho \cos \phi \sin \phi + (\rho \cos \phi + z) \sin \phi] \mathbf{a}_\rho \\ &+ [-\rho \sin^2 \phi + (\rho \cos \phi + z) \cos \phi] \mathbf{a}_\phi \end{aligned}$$

At P

$$\rho = \sqrt{40}, \quad \tan \phi = \frac{6}{-2}$$

ELECTRO MAGNETIC THEORY

Coordinate

Hence,

$$\begin{aligned}\cos \phi &= \frac{-2}{\sqrt{40}}, \quad \sin \phi = \frac{6}{\sqrt{40}} \\ \mathbf{A} &= \left[\sqrt{40} \cdot \frac{-2}{\sqrt{40}} \cdot \frac{6}{\sqrt{40}} + \left(\sqrt{40} \cdot \frac{-2}{\sqrt{40}} + 3 \right) \cdot \frac{6}{\sqrt{40}} \right] \mathbf{a}_\rho \\ &\quad + \left[-\sqrt{40} \cdot \frac{36}{40} + \left(\sqrt{40} \cdot \frac{-2}{\sqrt{40}} + 3 \right) \cdot \frac{-2}{\sqrt{40}} \right] \mathbf{a}_\phi \\ &= \frac{-6}{\sqrt{40}} \mathbf{a}_\rho - \frac{38}{\sqrt{40}} \mathbf{a}_\phi = -0.9487 \mathbf{a}_\rho - 6.008 \mathbf{a}_\phi\end{aligned}$$

Similarly, in the spherical system

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \begin{bmatrix} y \\ x + z \\ 0 \end{bmatrix}$$

or

$$A_r = y \sin \theta \cos \phi + (x + z) \sin \theta \sin \phi$$

$$A_\theta = y \cos \theta \cos \phi + (x + z) \cos \theta \sin \phi$$

$$A_\phi = -y \sin \phi + (x + z) \cos \phi$$

ELECTRO MAGNETIC THEORY

Coordinate

Systems



But $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, and $z = r \cos \theta$. Substituting these yields

$$\begin{aligned}\mathbf{A} &= (A_r, A_\theta, A_\phi) \\ &= r[\sin^2 \theta \cos \phi \sin \phi + (\sin \theta \cos \phi + \cos \theta) \sin \theta \sin \phi]\mathbf{a}_r \\ &\quad + r[\sin \theta \cos \theta \sin \phi \cos \phi + (\sin \theta \cos \phi + \cos \theta) \cos \theta \sin \phi]\mathbf{a}_\theta \\ &\quad + r[-\sin \theta \sin^2 \phi + (\sin \theta \cos \phi + \cos \theta) \cos \phi]\mathbf{a}_\phi\end{aligned}$$

At P

$$r = 7, \quad \tan \phi = \frac{6}{-2}, \quad \tan \theta = \frac{\sqrt{40}}{3}$$

ELECTRO MAGNETIC THEORY

Coordinate

Systems



Hence,

$$\begin{aligned}\cos \phi &= \frac{-2}{\sqrt{40}}, & \sin \phi &= \frac{6}{\sqrt{40}}, & \cos \theta &= \frac{3}{7}, & \sin \theta &= \frac{\sqrt{40}}{7} \\ \mathbf{A} &= 7 \cdot \left[\frac{40}{49} \cdot \frac{-2}{\sqrt{40}} \cdot \frac{6}{\sqrt{40}} + \left(\frac{\sqrt{40}}{7} \cdot \frac{-2}{\sqrt{40}} + \frac{3}{7} \right) \cdot \frac{\sqrt{40}}{7} \cdot \frac{6}{\sqrt{40}} \right] \mathbf{a}_r \\ &+ 7 \cdot \left[\frac{\sqrt{40}}{7} \cdot \frac{3}{7} \cdot \frac{6}{\sqrt{40}} \cdot \frac{-2}{\sqrt{40}} + \left(\frac{\sqrt{40}}{7} \cdot \frac{-2}{\sqrt{40}} + \frac{3}{7} \right) \cdot \frac{3}{7} \cdot \frac{6}{\sqrt{40}} \right] \mathbf{a}_\theta \\ &+ 7 \cdot \left[\frac{-\sqrt{40}}{7} \cdot \frac{36}{40} + \left(\frac{\sqrt{40}}{7} \cdot \frac{-2}{\sqrt{40}} + \frac{3}{7} \right) \cdot \frac{-2}{\sqrt{40}} \right] \mathbf{a}_\phi \\ &= \frac{-6}{7} \mathbf{a}_r - \frac{18}{7\sqrt{40}} \mathbf{a}_\theta - \frac{38}{\sqrt{40}} \mathbf{a}_\phi \\ &= -0.8571 \mathbf{a}_r - 0.4066 \mathbf{a}_\theta - 6.008 \mathbf{a}_\phi\end{aligned}$$

Note that $|\mathbf{A}|$ is the same in the three systems; that is,

$$|\mathbf{A}(x, y, z)| = |\mathbf{A}(\rho, \phi, z)| = |\mathbf{A}(r, \theta, \phi)| = 6.083$$

ELECTRO MAGNETIC THEORY

Coordinate (Assignment

1. Two uniform vector fields are given by $\mathbf{E} = -5\mathbf{a}_\rho + 10\mathbf{a}_\phi + 3\mathbf{a}_z$ and $\mathbf{F} = \mathbf{a}_\rho + 2\mathbf{a}_\phi - 6\mathbf{a}_z$. Calculate

(a) $|\mathbf{E} \times \mathbf{F}|$

(b) The vector component of $\mathbf{E}$ at $P(5, \pi/2, 3)$ parallel to the line $x = 2, z = 3$

(c) The angle $\mathbf{E}$ makes with the surface $z = 3$ at P

2. Given the vector field

$$\mathbf{H} = \rho z \cos \phi \mathbf{a}_\rho + e^{-2} \sin \frac{\phi}{2} \mathbf{a}_\phi + \rho^2 \mathbf{a}_z$$

At point $(1, \pi/3, 0)$, find

(a) $\mathbf{H} \cdot \mathbf{a}_x$

(b) $\mathbf{H} \times \mathbf{a}_\theta$

(c) The vector component of $\mathbf{H}$ normal to surface $\rho = 1$

(d) The scalar component of $\mathbf{H}$ tangential to the plane $z = 0$

ELECTRO MAGNETIC THEORY

Coordinate Systems (Assignment)



Systems

3. Given a vector field

$$\mathbf{D} = r \sin \phi \mathbf{a}_r - \frac{1}{r} \sin \theta \cos \phi \mathbf{a}_\theta + r^2 \mathbf{a}_\phi$$

determine

- (a) $\mathbf{D}$ at $P(10, 150^\circ, 330^\circ)$
- (b) The component of $\mathbf{D}$ tangential to the spherical surface $r = 10$ at P
- (c) A unit vector at P perpendicular to $\mathbf{D}$ and tangential to the cone $\theta = 150^\circ$

4. If $\mathbf{A} = 3\mathbf{a}_r + 2\mathbf{a}_\theta - 6\mathbf{a}_\phi$ and $\mathbf{B} = 4\mathbf{a}_r + 3\mathbf{a}_\phi$, determine

- (a) $\mathbf{A} \cdot \mathbf{B}$
- (b) $|\mathbf{A} \times \mathbf{B}|$
- (c) The vector component of $\mathbf{A}$ along $\mathbf{a}_z$ at $(1, \pi/3, 5\pi/4)$

ELECTRO MAGNETI C

THEORY

Gradient , Divergence and Curl Concepts

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ELECTRO MAGNETIC THEORY

Gradient



The mathematical operation to find the rate of change in a certain direction is called gradient.

Now, the gradient of a scalar field is defined as:

$$\text{Gradient of } \phi = \text{grad } \phi = \frac{d\phi}{dN} \mathbf{a}_N$$

ELECTRO MAGNETIC THEORY

Gradient Contd. ,



Introducing the vector operator for gradient:

$$\nabla = \frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z$$

The above operator is called Del Operator .

Del operator applied directly on a scalar quantity is called Gradient

The output obtained from Gradient is

ELECTRO MAGNETIC THEORY

Gradient in all three

Coordinate Systems



Let V is a scalar function then
Gradient of V (Grad V) in all
coordinate system are as below

$$\nabla V = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z$$

Rectangular

$$\nabla V = \frac{\partial V}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi + \frac{\partial V}{\partial z} \mathbf{a}_z$$

Cylindrical

$$\nabla V = \frac{\partial V}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi$$

Spherical

ELECTRO MAGNETIC THEORY

Gradient



1

Find the gradient of the following scalar fields:

(a) $V = e^{-z} \sin 2x \cosh y$

(b) $U = \rho^2 z \cos 2\phi$

(c) $W = 10r \sin^2 \theta \cos \phi$

ELECTRO MAGNETIC THEORY

Gradient



1 (a) $\nabla V = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z$
 $= 2e^{-z} \cos 2x \cosh y \mathbf{a}_x + e^{-z} \sin 2x \sinh y \mathbf{a}_y - e^{-z} \sin 2x \cosh y \mathbf{a}_z$

(b) $\nabla U = \frac{\partial U}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial U}{\partial \phi} \mathbf{a}_\phi + \frac{\partial U}{\partial z} \mathbf{a}_z$
 $= 2\rho z \cos 2\phi \mathbf{a}_\rho - 2\rho z \sin 2\phi \mathbf{a}_\phi + \rho^2 \cos 2\phi \mathbf{a}_z$

(c) $\nabla W = \frac{\partial W}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial W}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial W}{\partial \phi} \mathbf{a}_\phi$
 $= 10 \sin^2 \theta \cos \phi \mathbf{a}_r + 10 \sin 2\theta \cos \phi \mathbf{a}_\theta - 10 \sin \theta \sin \phi \mathbf{a}_\phi$

ELECTRO MAGNETIC THEORY

Divergence



Divergence is an operation on a vector yielding a scalar, just like the dot product.

We define the del operator as a vector operator:

$$\nabla = \frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z$$

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Divergence Contd.,



Let $\mathbf{D}$ is a Vector represented by (D_x, D_y, D_z) or $D_x \mathbf{a}_x + D_y \mathbf{a}_y + D_z \mathbf{a}_z$

$$\nabla \cdot \mathbf{D} = \left(\frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z \right) \cdot (D_x \mathbf{a}_x + D_y \mathbf{a}_y + D_z \mathbf{a}_z)$$

$$\nabla \cdot \mathbf{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$

$$\text{div } \mathbf{D} = \nabla \cdot \mathbf{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$

ELECTRO MAGNETIC THEORY

Divergence in Cylindrical and Spherical Coordinate System

Cylindrical

Coordinate System

$$\nabla \cdot \mathbf{D} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho D_{\rho}) + \frac{1}{\rho} \frac{\partial D_{\phi}}{\partial \phi} + \frac{\partial D_z}{\partial z}$$

Spherical Coordinate

System

$$\nabla \cdot \mathbf{D} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 D_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta D_{\theta}) + \frac{1}{r \sin \theta} \frac{\partial D_{\phi}}{\partial \phi}$$

ELECTRO MAGNETIC THEORY



Divergence in Cylindrical and Spherical Coordinate System

2. Determine the divergence of these vector fields:

(a) $\mathbf{P} = x^2yz \mathbf{a}_x + xz \mathbf{a}_z$

(b) $\mathbf{Q} = \rho \sin \phi \mathbf{a}_\rho + \rho^2 z \mathbf{a}_\phi + z \cos \phi \mathbf{a}_z$

(c) $\mathbf{T} = \frac{1}{r^2} \cos \theta \mathbf{a}_r + r \sin \theta \cos \phi \mathbf{a}_\theta + \cos \theta \mathbf{a}_\phi$

ELECTRO MAGNETIC THEORY

Divergence in Cylindrical and Spherical Coordinate System

$$\begin{aligned} 2. \quad (a) \quad \nabla \cdot \mathbf{P} &= \frac{\partial}{\partial x} P_x + \frac{\partial}{\partial y} P_y + \frac{\partial}{\partial z} P_z \\ &= \frac{\partial}{\partial x} (x^2 y z) + \frac{\partial}{\partial y} (0) + \frac{\partial}{\partial z} (x z) \\ &= 2xyz + x \end{aligned}$$

$$\begin{aligned} (b) \quad \nabla \cdot \mathbf{Q} &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho Q_\rho) + \frac{1}{\rho} \frac{\partial}{\partial \phi} Q_\phi + \frac{\partial}{\partial z} Q_z \\ &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho^2 \sin \phi) + \frac{1}{\rho} \frac{\partial}{\partial \phi} (\rho^2 z) + \frac{\partial}{\partial z} (z \cos \phi) \\ &= 2 \sin \phi + \cos \phi \end{aligned}$$

$$\begin{aligned} (c) \quad \nabla \cdot \mathbf{T} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 T_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (T_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (T_\phi) \\ &= \frac{1}{r^2} \frac{\partial}{\partial r} (\cos \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (r \sin^2 \theta \cos \phi) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (\cos \theta) \\ &= 0 + \frac{1}{r \sin \theta} 2r \sin \theta \cos \theta \cos \phi + 0 \\ &= 2 \cos \theta \cos \phi \end{aligned}$$

ELECTRO MAGNETIC THEORY

Cur I



Rect angular
Coor di nat e Syst em

=

Cyl i ndr i cal Coor di nat e
Syst em =

ELECTRO MAGNETIC THEORY

Curl and Divergence ,



Spherical Coordinate System

=

If Curl of a vector is zero then it is called irrotational field.

If Divergence of a Vector is zero, it is called solenoidal field.

ELECTRO MAGNETIC THEORY

Divergence and Curl (Assignment)



1. Determine the divergence of the following vector fields and evaluate them at the specified points.
 - (a) $\mathbf{A} = yz\mathbf{a}_x + 4xy\mathbf{a}_y + y\mathbf{a}_z$ at $(1, -2, 3)$
 - (b) $\mathbf{B} = \rho z \sin \phi \mathbf{a}_\rho + 3\rho z^2 \cos \phi \mathbf{a}_\phi$ at $(5, \pi/2, 1)$
 - (c) $\mathbf{C} = 2r \cos \theta \cos \phi \mathbf{a}_r + r^{1/2} \mathbf{a}_\phi$ at $(1, \pi/6, \pi/3)$
2. Given $\Phi = xy + yz + xz$, find gradient Φ at point $(1, 2, 3)$ and the directional derivative of Φ at the same point in the direction toward point $(3, 4, 4)$.
3. Find the divergence and curl of the following vectors:
 - (a) $\mathbf{A} = e^{xy} \mathbf{a}_x + \sin xy \mathbf{a}_y + \cos^2 xz \mathbf{a}_z$
 - (b) $\mathbf{B} = \rho z^2 \cos \phi \mathbf{a}_\rho + z \sin^2 \phi \mathbf{a}_z$
 - (c) $\mathbf{C} = r \cos \theta \mathbf{a}_r - \frac{1}{r} \sin \theta \mathbf{a}_\theta + 2r^2 \sin \theta \mathbf{a}_\phi$

ELECTRO MAGNETIC THEORY

Gradient, Divergence and Curl (Assignment)



1. **Answer:** (a) $4x, 4$, (b) $(2 - 3z)z \sin \phi, -1$, (c) $6 \cos \theta \cos \phi, 2.598$.

2. **Answer:** $5\mathbf{a}_x + 4\mathbf{a}_y + 3\mathbf{a}_z, 7$.



THANK YOU

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